Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C/C++, SQL, JavaScript
Software Development: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, FastAPI, Flask, React.js, Node.js, HTML, CSS
Databases: MySQL, SQLite, MongoDB, Redis, Database Design, SQL Queries, Data Modeling
AI & Machine Learning: Scikit-learn, PyTorch, TensorFlow/Keras, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Google Gemini API, EasyOCR
Analytics & Visualization: Pandas, NumPy, Power BI, DAX, Exploratory Data Analysis, Statistical Analysis
Engineering & Tools: Git/GitHub, PyTest, Unit Testing, Debugging, Test-Driven Development (TDD), Jupyter Notebook, VS Code

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack receipt processing platform that extracted structured data from 12,000+ receipts with approximately 96% OCR accuracy using EasyOCR and OpenCV.
- Developed secure FastAPI backend services with JWT authentication, Google OAuth, and REST APIs, reducing average API response time to under 180 ms.
- Automated expense categorization and analytics through NLP-based processing, reducing manual bookkeeping effort by 80% while providing interactive financial dashboards.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value store in Python, implementing a custom RESP protocol parser and multithreaded TCP server for concurrent client requests.
- Implemented Write-Ahead Logging, crash recovery, thread-safe storage, and LRU caching, combining systems programming with performance-oriented data structures.
- Validated reliability through 46 automated tests and benchmarked 619K SET operations/sec, demonstrating correctness and high-throughput request handling.

***Docwise* |LINK**

- Engineered an end-to-end RAG-based document analytics pipeline using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Developed the complete document intelligence workflow from document extraction and chunking to retrieval, contextual ranking, and LLM-based response generation.
- Improved response reliability through retrieval tuning, prompt engineering, and hallucination-aware validation, increasing contextual relevance and answer consistency.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025